

Week 5 DBMS Task

Payment Transaction Management System

1. Objective

To design and implement a Payment Transaction Management System that stores payment details, tracks transaction status, analyzes payment methods, and generates transaction reports.

2. Requirements

1. Create a Payment table.
2. Store payment details such as Payment ID, Customer ID, Payment Mode, Payment Date, Amount, and Payment Status.
3. Manage successful and failed transactions.
4. Analyze the different payment methods used by customers.
5. Generate payment transaction reports.

3. Database Design

Column	Data Type	Constraint
Payment_ID	INT	Primary Key
Customer_ID	INT	Foreign Key / NOT NULL
Payment_Mode	VARCHAR(30)	NOT NULL
Payment_Date	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL
Payment_Status	VARCHAR(20)	NOT NULL

4. SQL Implementation

```
CREATE DATABASE PaymentDB;
```

```
USE PaymentDB;
```

```
CREATE TABLE Payment (  
    Payment_ID INT PRIMARY KEY,  
    Customer_ID INT NOT NULL,  
    Payment_Mode VARCHAR(30) NOT NULL,  
    Payment_Date DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL,  
    Payment_Status VARCHAR(20) NOT NULL  
);
```

5. Insert Sample Records

```
INSERT INTO Payment  
(Payment_ID, Customer_ID, Payment_Mode, Payment_Date, Amount, Payment_Status)  
VALUES
```

```
(101, 1, 'UPI', '2026-08-01', 1500.00, 'Successful'),
(102, 2, 'Credit Card', '2026-08-02', 2500.00, 'Successful'),
(103, 3, 'Debit Card', '2026-08-03', 1200.00, 'Failed'),
(104, 1, 'Net Banking', '2026-08-04', 3000.00, 'Successful'),
(105, 4, 'UPI', '2026-08-05', 800.00, 'Failed'),
(106, 5, 'Credit Card', '2026-08-06', 4500.00, 'Successful');
```

6. Display All Payments

```
SELECT * FROM Payment;
```

7. Display Successful Transactions

```
SELECT *
FROM Payment
WHERE Payment_Status = 'Successful';
```

8. Display Failed Transactions

```
SELECT *
FROM Payment
WHERE Payment_Status = 'Failed';
```

9. Analyze Payment Methods

```
SELECT Payment_Mode, COUNT(*) AS Total_Transactions
FROM Payment
GROUP BY Payment_Mode;
```

10. Calculate Total Successful Payment Amount

```
SELECT SUM(Amount) AS Total_Successful_Amount
FROM Payment
WHERE Payment_Status = 'Successful';
```

11. Generate Payment Transaction Report

```
SELECT
    Payment_Mode,
    Payment_Status,
    COUNT(*) AS Transaction_Count,
    SUM(Amount) AS Total_Amount
FROM Payment
GROUP BY Payment_Mode, Payment_Status
ORDER BY Payment_Mode;
```

12. Find the Highest Payment

```
SELECT *
FROM Payment
ORDER BY Amount DESC
LIMIT 1;
```

13. Conclusion

The Payment Transaction Management System successfully stores and manages payment information. SQL queries can be used to identify successful and failed transactions, analyze customer payment methods, calculate transaction amounts, and generate useful payment reports.